Proposal: Automated Regulatory Submission Assistant

Prepared by: Kvant Consulting **For:** Krka, d.d., Novo Mesto — Regulatory Affairs Division **Date:** 2023-09-04 **Version:** 1.2 (revised after scoping call August 28) **Contact:** Anja Štrukelj (anja@kvant.si), Žiga Potočnik (ziga@kvant.si)

Executive Summary

Krka's Regulatory Affairs team currently manages submission dossiers for 40+ markets simultaneously. The manual process of cross-referencing CTD (Common Technical Document) requirements by market, tracking submission status, and drafting cover letters consumes an estimated 6,200 hours of skilled regulatory staff time annually.

Kvant proposes a two-phase AI implementation: (1) a retrieval-augmented assistant for querying regulatory guidelines and Krka's own submission history, and (2) a drafting assistant for templated cover letters and gap analyses. This proposal covers Phase 1 only.

Total proposed budget, Phase 1: **€62,000** Estimated delivery: 14 weeks from project start

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Problem Statement

Regulatory Affairs staff at Krka face three recurring bottlenecks:

1. Guideline retrieval is slow and fragmented. EMA, FDA, ICH, and local health authority guidelines are stored across multiple external portals and Krka's internal SharePoint. There is no unified search. Staff routinely spend 30–90 minutes locating the applicable guideline for a specific product/market combination.

2. Submission history is underutilized. Krka has 20+ years of internal submission dossiers and correspondence. This institutional knowledge is practically inaccessible — the SharePoint search returns full documents, not relevant passages.

3. Cover letter drafting is repetitive but error-prone. Standard sections are copy-pasted from previous submissions with manual adjustments. Errors in market-specific language have caused minor compliance issues in two cases in the past three years (details not shared with us, but flagged by the RA Director during scoping).

Proposed Solution: Phase 1

Architecture

We propose a retrieval-augmented generation (RAG) system with the following components:

Document Corpus (ingestion): - Krka internal submission history: ~3,200 documents (PDF, DOCX, structured XML) - Applicable regulatory guidelines: EMA, ICH, selected national authority guidelines (~800 documents) - Updated quarterly via automated pipeline

Embedding and Indexing: - Chunking: hybrid approach — semantic chunking for narrative sections, fixed-size with overlap for tabular/structured content - Embeddings: `text-embedding-3-large` (OpenAI) with evaluation against Voyage AI embeddings for pharmaceutical/regulatory domain text - Vector store: Qdrant, self-hosted on dedicated Hetzner instance (CCX33, 32GB RAM)

Retrieval: - Hybrid retrieval: dense vector search + BM25 sparse retrieval, with reciprocal rank fusion - Metadata filtering by market, product category, guideline body, and date range - Re-ranking via cross-encoder (BGE reranker)

Generation: - Primary: GPT-4o with grounded prompting (citations mandatory in system prompt) - Fallback / cost optimization: GPT-4o-mini for simple lookups - "I don't know" behavior: explicit instruction + confidence threshold from retrieval scores

Evaluation (ongoing): - RAGAS framework for faithfulness, answer relevancy, context precision - Internal golden dataset: 60 Q&A pairs prepared by Krka RA staff during onboarding - LangSmith for tracing and debugging

Interface: - Web UI: FastAPI backend, simple React frontend - Deployment: Docker on Hetzner, internal Krka network access only - No SSO in Phase 1 (Krka IT confirmed this is acceptable)

Corpus Handling and Data Privacy

All documents remain on Krka-controlled infrastructure or Hetzner EU instances. No data is sent to OpenAI training pipelines (API usage with `store: false`). Embeddings are generated locally via OpenAI API call but vectors themselves are stored on-premises.

Krka's legal team has reviewed the OpenAI data processing agreement. Kvant recommends Krka countersign the DPA addendum before project start — we can facilitate.

Deliverables

#	Deliverable	Week
1	Data audit report + ingestion pipeline (Krka internal docs)	3
2	Ingestion pipeline (external regulatory guidelines)	5
3	RAG prototype (internal access, Kvant-hosted for testing)	7
4	Golden dataset QA with Krka RA team	9
5	Evaluation report (RAGAS + human review)	11
6	Production deployment on Hetzner + handover	14
7	Documentation + training session (half day, Novo Mesto)	14

Budget Breakdown

Item	Cost (EUR)
Discovery and data audit (8 days)	€8,800
Ingestion pipeline development	€12,000
RAG system development and iteration	€18,000
Evaluation framework + golden dataset	€6,500
Production deployment and DevOps	€7,200
Training and documentation	€3,500
Infrastructure (Hetzner, 6 months)	€2,400
Contingency (8%)	€3,600
Total	**€62,000**

Infrastructure costs transfer to Krka after 6 months (~€400/month ongoing).

OpenAI API costs are not included — estimated €800–1,400/month at projected query volumes (this estimate may be revised during discovery).

Team

- **Anja Štrukelj** — project lead, client liaison, evaluation design
- **Žiga Potočnik** — RAG architecture, embeddings, retrieval optimization
- **Nika Bregar** — ingestion pipeline, data engineering
- **Matic Demšar** — infrastructure and deployment
- **Maša Vidmar** — technical review and quality assurance

Risks and Mitigations

Risk	Likelihood	Mitigation
Document quality is poor (scanned PDFs, no OCR)	Medium	OCR pass in ingestion pipeline; cost TBD
Krka RA staff unavailable for golden dataset creation	Medium	Schedule during project scoping, not discovery
GPT-4o cost higher than projected	Low-Medium	GPT-4o-mini fallback for retrieval-only queries
EMA/ICH guidelines update mid-project	Low	Versioned ingestion; quarterly refresh pipeline
SharePoint access permissions are complex	Medium	Flagged for IT kickoff — must resolve in week 1

Assumptions

This proposal assumes: - Krka provides a dedicated IT contact for SharePoint/infrastructure coordination - Source documents are predominantly machine-readable (not scanned images) - Krka RA staff commit approximately 12 hours across the project for golden dataset review - Project start no later than October 7, 2023

Significant deviation from the above will require scope renegotiation.

Notes

This proposal supersedes the draft shared August 15, 2023 (v1.0), which incorrectly scoped the corpus size at ~1,500 documents. The revised estimate of 4,000 documents reflects the SharePoint audit Krka IT provided on August 27.

Phase 2 (cover letter drafting assistant) is not scoped here. We are happy to begin scoping discussions during Phase 1 delivery, with a view to a Phase 2 proposal by week 10.